

Bidirectional EN ↔ ES translation transformer

ENCODER

Reading the diagram

- data flow
- - - residual connection
- - - encoder to decoder

37.6M parameters
4 encoder + 4 decoder layers
d_model 512, 8 heads, d_ff 2048
one shared 16k vocabulary
embeddings tied 3 ways

the only path from source to target

memory
(6, 512)

K, V

4x

4x

source: <2es> I am tired . </s>

the direction tag tells one model which way to translate

DECODER

Estoy cansado . </s>

next-token distribution

$P(\text{token} \mid \text{source, target so far})$

Softmax

Linear (tied to embedding)

512 → 16,000

Add & LayerNorm

Feed-Forward Network

512 → 2048 → 512, ReLU

Add & LayerNorm

Multi-Head Cross-Attention

Q from decoder, K,V from encoder

Add & LayerNorm

Masked Multi-Head Self-Attention

cannot see the future

+ Positional Encoding

sinusoidal, fixed

Output Embedding

same shared table

target so far: <s> Estoy cansado .

shifted right: position t predicts token t+1